Executive Summary: Credit Risk ML Model

1. The Business Challenge

In consumer lending, optimizing a machine learning model for standard accuracy fails to capture the true financial impact. The cost of a False Negative (missing a GBP 25,000 defaulted loan) is significantly higher than a False Positive (wrongly rejecting a good customer, losing GBP 2,500 in lifetime value). This 10:1 cost asymmetry requires a specialized, cost-sensitive approach.

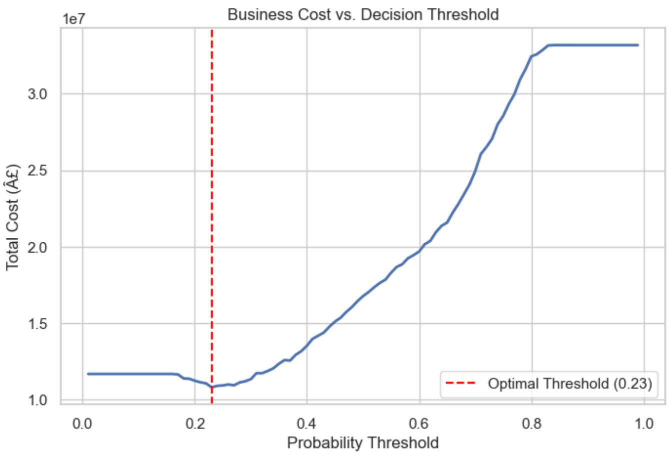
2. The Solution & Financial Impact

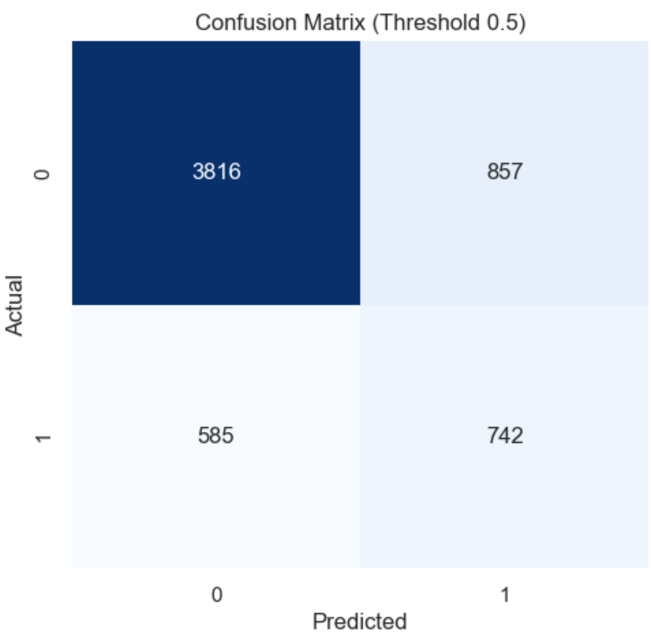
By simulating the financial costs across different probability thresholds using our custom XGBoost model, we computationally derived the optimal decision threshold: 0.23 (rather than the default 0.50).

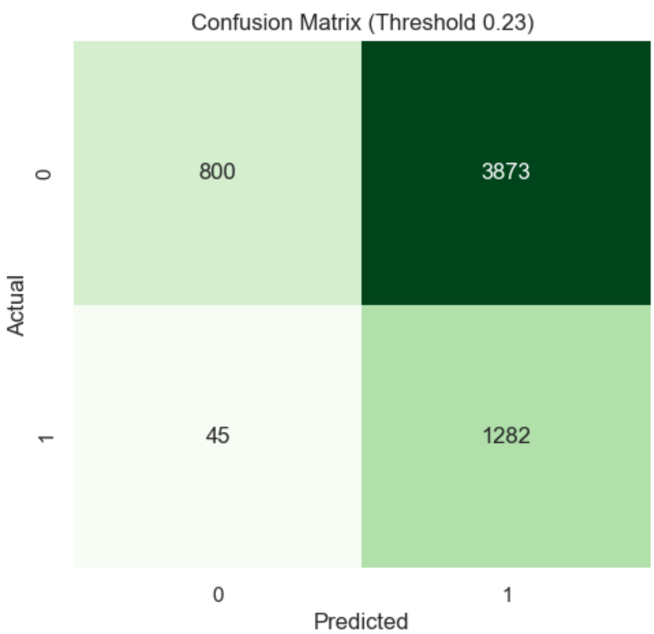
Business Impact on the 6,000-customer test set:

- We

3. Performance & Threshold Analysis







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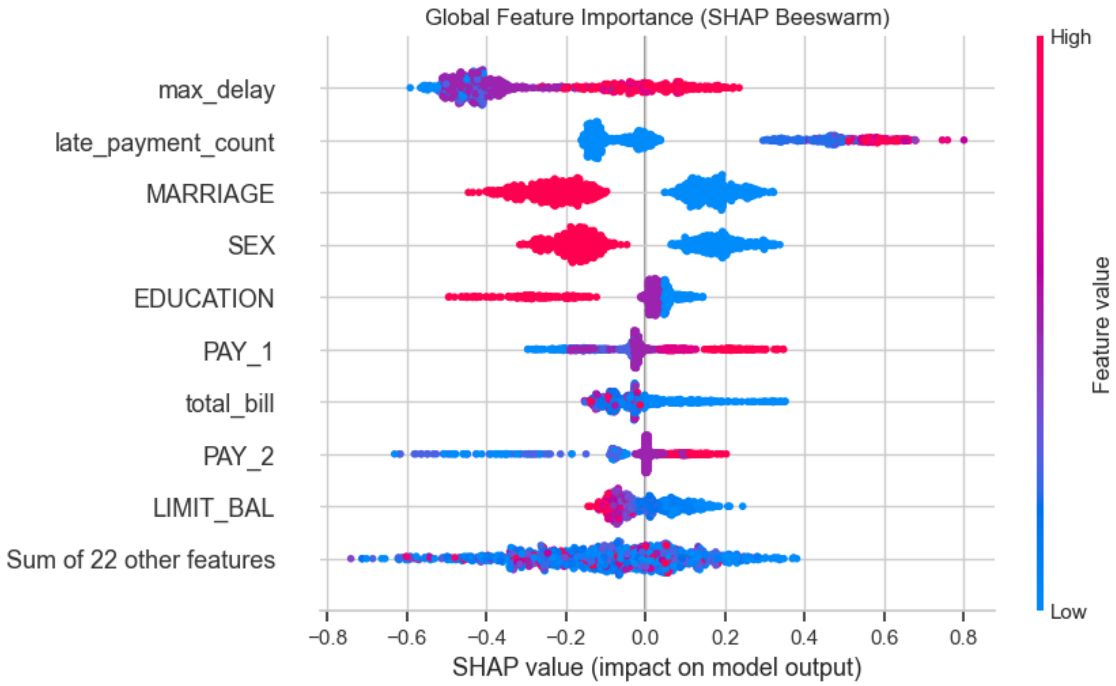
4. Explainability and Risk Factors

The model utilizes SHAP (SHapley Additive exPlanations) to provide full transparency. The beeswarm plot below shows the most critical risk drivers across the entire portfolio. Payment delays (PAY_1), low limit balances, and high utilization rates are the strongest indicators of default.



5. Conclusion & Recommendations

The cost-sensitive model successfully bridges the gap between technical metrics and business ROI. We recommend deploying this model in a shadow-testing environment to validate the projected GBP 5.96M savings per 6,000 applicants, while actively monitoring the top SHAP features for macroeconomic drift.